RA.SANJEY ADHITHYAA
192412144

COURSE OUTCOME-3
Design and Performance Evaluation of an Integrated Photodetection, Imaging, Biomedical Sensing and Display System

INTRODUCTION

Optoelectronic systems combine optical and electronic technologies to detect, process, and display information. These systems are widely used in biomedical monitoring, automotive safety, imaging, communication, and consumer electronics.

The proposed system integrates PIN and APD photodetectors, CMOS image sensors, photovoltaic devices, biomedical sensors, LiDAR technology, ultraviolet detection, and advanced display technologies.

The main objective is to develop an efficient optoelectronic platform with high sensitivity, low noise, fast response, low power consumption, and reliable performance.

STAGE 1 – INTERPRET THE APPLICATION REQUIREMENTS

1.1 Photodetection Requirements

The proposed system requires photodetectors capable of converting optical signals into electrical signals accurately.

Important requirements include:

1. High responsivity to detect small variations in optical power.
2. High quantum efficiency for effective photon conversion.
3. Low noise-equivalent power for weak-light detection.
4. High detectivity for improved sensitivity.
5. Fast response time for LiDAR and imaging.
6. Good linearity for accurate optical measurements.

7. Low power consumption for wearable applications.

Wearable PPG Requirements

Photoplethysmography (PPG) measures changes in blood volume using light.

A suitable PPG detector requires high sensitivity, low dark current, good linearity, and low power consumption.

The detector must accurately identify small changes in reflected optical intensity caused by blood circulation.

Automotive LiDAR Requirements

LiDAR uses laser pulses to measure distances.

The receiver requires:

- High-speed photodetection.
- High sensitivity to weak reflected signals.
- Low noise.
- High bandwidth.
- Accurate timing measurements.

An avalanche photodiode is suitable because its internal multiplication gain improves weaksignal detection.

Low-Light Detection Requirements

Low-light applications require detectors with low dark current, low NEP, high detectivity, and effective noise-reduction techniques.

Responsivity and Quantum Efficiency

Responsivity measures the electrical current generated per unit optical power.

$$R = I_p / P$$

Quantum efficiency represents the fraction of incident photons converted into collected charge carriers.

$\eta = (R \times h \times c) / (q \times \lambda)$ Where:

- R = Responsivity
- h = Planck's constant
- c = Speed of light

- q = Electron charge
- λ = Wavelength

NEP and Detectivity

Noise-equivalent power (NEP) is the optical input power required to produce a signal-tonoise ratio of one in a specified bandwidth.

A lower NEP indicates better sensitivity.

Specific detectivity is expressed as: D*
= √(A × Δf) / NEP

Where A is the detector area and Δf is the noise bandwidth.

Detector Linearity

Detector linearity ensures that photocurrent changes proportionally with incident optical power.

Good linearity is essential for accurate imaging, biomedical measurements, and optical sensing.

Response Time and Bandwidth

Response time determines how quickly a detector responds to changes in optical intensity.

Detector bandwidth is limited by carrier transit time and the RC time constant.

A smaller detector capacitance generally improves bandwidth.

CMOS and CCD Requirements

CMOS sensors provide low power consumption, fast readout, and easy integration with processing circuits.

CCD sensors provide high-quality imaging but generally require more complex readout circuitry.

Photovoltaic Requirements

Flexible photovoltaic devices should be lightweight, mechanically flexible, efficient, and suitable for wearable energy harvesting.

Display Requirements

OLED, MicroLED, and quantum-dot technologies should be evaluated based on brightness, contrast, power consumption, response speed, colour quality, and reliability.

APPLICATION REQUIREMENT STATEMENT

The proposed integrated optoelectronic platform must provide accurate biomedical sensing, high-speed LiDAR detection, reliable imaging, ultraviolet monitoring, and efficient display operation.

STAGE 2 – INVESTIGATE AND SELECT THE PHOTODETECTOR CONFIGURATION

2.1 Photocurrent Calculation for PPG

Given:

Responsivity:

$R = 0.60$ A/W

Optical-power variation:

$\Delta P = 25\ \mu W\ \Delta P$

$= 25 \times 10^{-6}$ W

Formula:

$\Delta I_p = R \times \Delta P$

Substitution:

$\Delta I_p = 0.60 \times 25 \times 10^{-6}$

$\Delta I_p = 15 \times 10^{-6}$ A

Final Answer:

$\Delta I_p = 15\ \mu A$

The PPG sensor produces a photocurrent variation of 15 μA.

This electrical signal represents changes in received light intensity associated with pulsatile blood-volume variations.

After filtering and processing, the signal can be used to estimate heart rate.

2.2 Comparison Between PIN and APD Photodetectors

Parameter	**PIN Photodiode**	**APD**
Responsivity	Moderate to high	High effective responsivity
Internal gain	Approximately 1	Typically greater than 1

Sensitivity	Good	Higher for suitable low-light conditions
Noise	Relatively low	Additional multiplication noise
Bandwidth	High	High, depending on design
Bias voltage	Low	Higher
Circuit complexity	Simple	More complex
Power consumption	Generally lower	Generally higher
Applications	PPG and imaging	LiDAR and weak-signal detection

2.3 LiDAR Photocurrent Calculation

Given:

Received optical power:

$P = 3\ \mu W$ $P =$

$3 \times 10^{-6}\ W$

Responsivity:

$R = 0.80\ A/W$

Formula:

$I_p = R \times P$

Substitution:

$I_p = 0.80 \times 3 \times 10^{-6}$

$I_p = 2.4 \times 10^{-6}\ A$

Final Answer:

$I_p = 2.4\ \mu A$

2.4 Dark Current

Dark current is the small current flowing through a photodetector even when no light is present.

It contributes to shot noise and reduces the ability to detect weak optical signals.

Dark current can be reduced by selecting low-leakage detectors, controlling temperature, and using suitable bias conditions.

2.5 Quantum Efficiency

Quantum efficiency determines the efficiency of photon-to-electron conversion.

Higher quantum efficiency generally improves the primary responsivity of a photodetector at a given wavelength.

2.6 NEP and Detectivity

NEP indicates the minimum detectable optical signal relative to detector noise.

Specific detectivity provides a normalized sensitivity measure that accounts for detector area and bandwidth.

For reliable low-light operation, a detector should have low NEP and high specific detectivity.

Numerical values cannot be determined without additional detector-noise, area, and bandwidth information.

DESIGN DECISION 1

I recommend a PIN photodiode for wearable PPG sensing and an avalanche photodiode for automotive LiDAR and weak-signal detection.

The PIN photodiode provides low noise, low operating voltage, and simple signal conditioning.

The APD provides internal multiplication gain, improving the detection of weak reflected optical signals when its gain and noise are properly optimized.

STAGE 3 – DEVELOP AND EVALUATE THE NOISE AND SNR STRATEGY

3.1 Signal-to-Noise Ratio Calculation

Given:

Given:

Signal power:

Ps = −22 dBm

Noise power: Pn

= −37 dBm

Formula:

SNR (dB) = Ps (dBm) − Pn (dBm)

Substitution:

SNR = −22 − (−37)

SNR = −22 + 37 **Final**

Answer:

SNR = 15 dB

Formula:

SNR (linear) = 10^(SNR / 10) **Substitution:**

SNR (linear) = 10^(15 / 10)

SNR (linear) = $10^{1.5}$ **Final**

Answer:

SNR (linear) = 31.62

The received signal power is approximately 31.62 times the noise power.

3.2 Types of Noise

Shot Noise

Shot noise occurs because electric charge is transferred in discrete units.

The RMS shot-noise current is:

$i_{shot} = \sqrt{(2 \times q \times I \times \Delta f)}$

Where:

- q = Electron charge
- I = Average current
- Δf = Noise bandwidth

Dark-Current Noise

Dark-current noise is caused by random fluctuations associated with detector leakage current.

It becomes especially important when the received optical signal is weak.

Avalanche Noise

Avalanche noise occurs in APDs because the multiplication process is random.

The APD shot-noise model can be expressed as:

3.5 Avalanche Noise

Avalanche noise occurs in avalanche photodiodes (APDs) due to the random multiplication of charge carriers during the avalanche process.

Formula:

$i_n^2 = 2 \times q \times (I_p + I_a) \times M^2 \times F(M) \times \Delta f$

Here, (I_d) is assumed to represent the primary multiplied dark current, and (F(M)) is the excess-noise factor.

Thermal Noise

Thermal noise is produced by the random movement of charge carriers in resistive circuit elements.

For a resistance R: $i_{thermal} =$

$\sqrt{(4 \times k \times T \times \Delta f / R)}$

3.3 Dominant Noise Under Low-Light Conditions

Under low-light conditions, dark-current noise and receiver electronic noise may become significant.

In APDs, multiplication noise must also be considered.

The dominant noise source depends on detector characteristics, operating temperature, gain, and receiver bandwidth.

3.4 Effect of Decreasing Optical Power

When received optical power decreases, photocurrent also decreases.

If dark current and electronic noise remain approximately constant, the SNR becomes lower.

This reduces the reliability of weak-signal detection.

3.5 Effect of Bandwidth

Increasing the receiver bandwidth generally increases integrated noise.

Reducing bandwidth can improve SNR, provided the required signal information is preserved.

3.6 Methods to Improve SNR

1. Use an appropriate APD multiplication gain.
2. Apply optical filters to reject unwanted wavelengths.
3. Select photodetectors with low dark current.
4. Use temperature control or cooling where necessary.
5. Optimize receiver bandwidth.
6. Improve optical alignment and coupling.
7. Use low-noise amplifiers.
8. Apply suitable digital filtering.

DESIGN DECISION 2

I recommend a low-noise receiver configuration using optimized APD gain, optical filtering, low-dark-current detectors, and bandwidth optimization.

The calculated SNR is 15 dB. This provides a useful starting point for weak-signal detection, but reliable operation must be confirmed against the required detection probability, falsealarm rate, and application-specific performance limits.

STAGE 4 – ANALYZE RESPONSE TIME, BANDWIDTH AND HIGH-SPEED DETECTION

4.1 Transit-Time Bandwidth Calculation

Given:

Transit time: t_r

= 20 ps

Converting into seconds:

$t_r = 20 \times 10^{-12}$ s

Formula:

$f = 0.44 / t_r$ **Substitution:**

$f = 0.44 / (20 \times 10^{-12})$

$f = 0.022 \times 10^{12}$ Hz f

$= 22 \times 10^{9}$ Hz **Final**

Answer: f = 22 GHz

The approximate transit-time-limited bandwidth is 22 GHz.

The actual receiver bandwidth may be lower because of capacitance, resistance, avalanche build-up time, and electronic circuit limitations.

4.2 Carrier Transit Time

Carrier transit time is the time required for generated charge carriers to travel across the detector's active region.

A shorter transit time generally supports faster photodetection.

4.3 RC Time Constant

The RC time constant also limits detector response. τ_{rc}

$= R \times C$

The corresponding RC-limited bandwidth is approximately:

$f_{rc} = 1 / (2 \times \pi \times R \times C)$

4.4 Effect of Detector Capacitance

Higher detector capacitance generally increases the RC time constant and reduces bandwidth.

Smaller detector areas often reduce capacitance, but they may also reduce the amount of collected light.

4.5 Detector Area and Sensitivity Trade-Off

A large detector area improves optical collection and alignment tolerance.

However, it generally increases capacitance and may reduce speed.

A smaller detector can achieve higher bandwidth but requires improved optical focusing and alignment.

4.6 Suitable High-Speed Detector

A small-area, high-speed APD with low-capacitance packaging and a suitable transimpedance amplifier is recommended for weak-return LiDAR applications.

A CMOS image sensor is recommended for high-speed image acquisition because it supports parallel readout and on-chip signal processing.

DESIGN DECISION 3

I recommend a small-area, low-capacitance APD receiver with optimized optical coupling and high-speed amplification.

The calculated transit-time bandwidth is 22 GHz, indicating a high theoretical speed capability. The complete receiver must still be tested to verify its practical bandwidth.

STAGE 5 – EVALUATE PV, IMAGE-SENSOR AND EMERGING DISPLAY TECHNOLOGIES

A. SOLAR PHOTOVOLTAIC ANALYSIS

5.1 Maximum Power Calculation

Given:

Open-circuit voltage:

Voc = 0.60 V

Short-circuit current:

Isc = 5 A

Fill factor:

FF = 0.75

Formula:

P_{max} = Voc × Isc × FF **Substitution:**

P_{max} = 0.60 × 5 × 0.75

P_{max} = 3 × 0.75

P_{max} = 2.25 W

Final Answer:

P_{max} = 2.25 W

The maximum power output of the specified photovoltaic device is 2.25 W under the operating conditions represented by the given parameters.

5.2 Open-Circuit Voltage

Open-circuit voltage is the maximum terminal voltage when no external current flows.

5.3 Short-Circuit Current

Short-circuit current is the current delivered when the PV terminals are short-circuited. **5.4**

Fill Factor

Fill factor indicates the quality of the photovoltaic I–V characteristic.

FF = P_{max} / (Voc × Isc)

A higher fill factor generally indicates better electrical performance.

5.5 PV Efficiency

$\eta PV = (P_{max} / P_{inci}d_{ent}) \times 100\%$

Numerical efficiency cannot be calculated because the incident optical power is not provided.

5.6 Flexible Photovoltaic Devices

Flexible PV devices can be integrated into wearable electronics because they are lightweight and can conform to curved surfaces.

Advantages:

- Lightweight construction.
- Mechanical flexibility.
- Auxiliary energy harvesting.
- Reduced dependence on external charging.

Limitations:

- Limited output under indoor lighting.
- Mechanical durability concerns.
- Variable power generation.
- Possible lower efficiency than conventional rigid devices.

For wearable operation, flexible PV is best used with a rechargeable battery and powermanagement circuit.

B. CMOS AND CCD IMAGE-SENSOR ANALYSIS

5.7 Comparison Between CMOS and CCD

Parameter	**CMOS**	**CCD**
Architecture	Active-pixel sensor	Charge-transfer architecture
Pixel operation	Local charge-to-voltage conversion	Charge accumulation and transfer
Readout	Column/parallel readout	Sequential charge transfer

Power consumption	Generally lower	Generally higher
Speed	High	Depends on architecture
Noise	Low in modern designs	Traditionally low-noise imaging
Pixel scaling	Strong integration capability	More specialized implementation
SoC integration	Excellent	More difficult
Face recognition	Highly suitable	Suitable but less common
Automotive imaging	Highly suitable	Less commonly selected

5.8 Pixel Scaling

Pixel scaling refers to reducing the physical dimensions of image-sensor pixels.

Smaller pixels allow higher image resolution within a compact sensor area.

However, reducing pixel size can decrease photon collection and full-well capacity.

Modern technologies such as backside illumination and advanced pixel structures help improve performance.

5.9 SoC Integration

CMOS technology allows image sensing, amplification, analog-to-digital conversion, and processing functions to be integrated efficiently.

This reduces circuit complexity and supports compact electronic devices.

Image-Sensor Selection

A CMOS image sensor is recommended because it provides low power consumption, highspeed readout, compact integration, and compatibility with face-recognition and automotive applications.

C. EMERGING DISPLAY ANALYSIS

5.10 OLED, MicroLED and Quantum-Dot Comparison

Parameter	**OLED**	**MicroLED**	**Quantum-Dot Display**

Light generation	Organic electroluminescence	Inorganic LED emission	Quantum-dot conversion or emission
Brightness	High	Very high potential	High, depending on architecture
Contrast	Excellent	Excellent	Architecture-dependent
Response speed	Very fast	Very fast	Depends on display architecture
Efficiency	Good	High potential	Technology-dependent
Colour performance	Excellent	Excellent	Excellent colour gamut
Lifetime	Organic ageing possible	Long potential lifetime	Depends on implementation
Power consumption	Content-dependent	Potentially efficient	Architecture-dependent
AR/VR/MR	Suitable	Highly promising	Suitable in selected designs

Quantum-dot displays include different architectures, such as quantum-dot-enhanced LCDs, quantum-dot OLEDs, and emerging electroluminescent quantum-dot displays. Their performance is therefore not identical.

5.11 OLED Technology

OLED displays use organic materials that emit light when electrical current passes through them.

They offer high contrast, thin construction, and flexible display capability.

5.12 MicroLED Technology

MicroLED displays use microscopic inorganic light-emitting diodes.

They offer high brightness, fast response, and potentially high efficiency and long lifetime.

5.13 Quantum-Dot Technology

Quantum dots are semiconductor nanocrystals with wavelength-dependent optical properties.

They support excellent colour reproduction and wide colour gamut.

DESIGN DECISION 4

I recommend flexible photovoltaic devices for auxiliary wearable power, CMOS image sensors for imaging, and OLED technology for the primary compact display.

Flexible PV supports energy harvesting, CMOS enables low-power integration, and OLED provides high contrast with practical flexible-display capability.

MicroLED is a promising alternative for future high-brightness AR applications.

STAGE 6 – EVALUATE BIOMEDICAL AND INDUSTRIAL APPLICATIONS

6.1 Wearable Biomedical Sensing

Photoplethysmography is a non-invasive optical technique used to monitor changes in blood volume.

An LED illuminates the skin, and a photodiode detects the reflected or transmitted light.

Blood-volume variations cause changes in detected optical intensity.

The photodiode converts these changes into electrical signals.

A signal-conditioning circuit amplifies and filters the signal before processing.

Proposed Configuration

- Optical source: Green LED for reflective wrist PPG.
- Detector: Silicon PIN photodiode.
- Responsivity: 0.60 A/W, as specified.
- Photocurrent variation: 15 μA.
- Signal conditioning: Transimpedance amplifier and filters.
- Processing: Microcontroller-based signal analysis.
- Power supply: Rechargeable battery with optional PV support.

6.2 NIR Spectroscopy

Near-infrared spectroscopy uses NIR light to investigate optical absorption and scattering in biological materials.

Different biological components interact differently with selected wavelengths.

The received optical signal is measured and processed to obtain information about the material.

Proposed Configuration

- NIR LED or laser source.
- Wavelength-compatible photodiode.
- Optical filter.
- Low-noise amplifier.
- Analog-to-digital converter.
- Digital signal-processing unit.

Wavelength selection depends on the target biological material and measurement objective.

Silicon photodiodes are suitable for part of the NIR region, while extended-wavelength applications may require InGaAs detectors.

6.3 Automotive LiDAR

LiDAR measures distance using the travel time of reflected laser light.

A laser pulse is transmitted toward an object.

The reflected signal is detected by a high-speed photodetector.

Distance is calculated using:

$d = (c \times t) / 2$

Where:

- d = Distance
- c = Speed of light
- t = Round-trip travel time

Proposed Configuration

- Pulsed laser transmitter.
- Optical focusing lens.
- APD receiver.
- Low-noise amplifier.
- Timing measurement circuit.
- Digital processing unit.

The APD provides sensitivity to weak optical returns, while the timing circuit supports accurate distance estimation.

The selected detector material must be compatible with the LiDAR wavelength.

6.4 Face Recognition

Face-recognition systems use image sensors to capture facial information.

CMOS image sensors are suitable because they provide high-speed image acquisition and efficient integration.

Near-infrared illumination can improve image capture under low-light conditions.

Proposed Configuration

- CMOS image sensor.
- NIR illumination source.
- Optical lens.
- Image-processing processor.
- Face-recognition algorithm.

The system must provide sufficient resolution, controlled illumination, and reliable imageprocessing performance.

6.5 Ultraviolet Sensing

UV sensors detect ultraviolet radiation for environmental monitoring, industrial safety, and exposure measurement.

Suitable UV detectors require:

1. High sensitivity within the required UV wavelength range.
2. Low dark current.
3. Good stability.
4. Low noise.
5. Strong rejection of unwanted visible light.
6. Suitable calibration.

A UV-enhanced silicon photodiode or a wide-bandgap semiconductor detector may be selected depending on the required spectral response.

DESIGN DECISION 5

I recommend an application-specific sensing configuration consisting of a PIN photodiode for PPG, wavelength-compatible photodiodes for NIR spectroscopy, an APD for LiDAR, a

CMOS image sensor for face recognition, and a UV-selective photodetector for ultraviolet monitoring.

This configuration balances sensitivity, speed, power consumption, and integration requirements.

STAGE 7 – SIMULATION AND PERFORMANCE VERIFICATION USING MATLAB

7.1 Aim

To simulate photodetector responsivity, noise, SNR, and bandwidth characteristics using MATLAB.

7.2 Input Parameters

Parameter	**Value**
PPG responsivity	0.60 A/W
PPG optical-power variation	25 μW
LiDAR responsivity	0.80 A/W
LiDAR optical power	3 μW
Signal power	−22 dBm
Noise power	−37 dBm
Transit time	20 ps
Assumed APD gain	10

7.3 MATLAB Program

```
clc; clear;
close all;
% Given parameters
R = 0.60;
P = linspace(0,50e-6,100);
```

```
% Responsivity analysis I = R .* P;
figure;
plot(P*1e6,I*1e6,'LineWidth',2);
xlabel('Optical Power (uW)');
ylabel('Photocurrent (uA)');
title('Photocurrent vs Optical Power');
grid on; % Noise analysis q = 1.602e-
19;
B = 1e6;

Id = 1e-9; shot_noise =
sqrt(2*q.*(I+Id)*B); dark_noise =
sqrt(2*q*Id*B); M = 10; k = 0.02; F =
k*M + (1-k)*(2-1/M); apd_noise =
M*sqrt(F).*shot_noise; figure;
plot(P*1e6,shot_noise*1e9,...
'LineWidth',2); hold on;
plot(P*1e6,apd_noise*1e9,...
'LineWidth',2); xlabel('Optical Power
(uW)'); ylabel('RMS Noise Current
(nA)'); title('PIN and APD Noise
Comparison'); legend('PIN Shot
Noise',...        'APD Multiplied Shot
Noise'); grid on;
% SNR analysis
% Illustrative shot-noise-limited model
snr_linear = I.^2 ./ ...
(2*q.*(I+Id)*B); snr_db =
10*log10(snr_linear); figure;
plot(P(2:end)*1e6,...
    snr_db(2:end),'LineWidth',2);
xlabel('Optical Power (uW)');
ylabel('SNR (dB)'); title('SNR
vs Optical Power'); grid on; G =
1:30; Ip = 0.8*3e-6;
Fgain = k.*G + ...        (1-
k).*(2-1./G); thermal_noise =
2e-9; noise_var = ...
   2*q*(Ip+Id)*B.*G.^2.*Fgain ...
+ thermal_noise^2; snr_gain =
(G.*Ip).^2 ./ noise_var; figure;
plot(G,10*log10(snr_gain),...
    'LineWidth',2);
xlabel('APD Gain');
ylabel('SNR (dB)');
```

```
title('SNR vs APD Gain');
grid on; tr = 20e-12; f =
0.44/tr;
% Theoretical results
PPG_current = 0.60*25e-6;
LiDAR_current = 0.80*3e-6;
SNR_given = -22-(-37); fprintf('PPG
Current = %.2f uA\n',...
PPG_current*1e6); fprintf('LiDAR
Current = %.2f uA\n',...
LiDAR_current*1e6); fprintf('Given
SNR = %.2f dB\n',...
SNR_given); fprintf('Bandwidth =
%.2f GHz\n',...
    f/1e9); fprintf('Dark Noise =
%.3f nA\n',...
    dark_noise*1e9);
```

Simulation Assumptions

The simulation uses an illustrative noise bandwidth of 1 MHz, dark current of 1 nA, APD ionization-ratio parameter of 0.02, and receiver thermal-noise current of 2 nA.

These values are assumed for modelling and are not specified in the assignment.

The SNR plots are based on detector-current noise models. They should not be treated as a direct reproduction of the separately specified −22 dBm signal and −37 dBm noise conditions.

7.4 Expected Simulation Results

Graph 1: Photocurrent vs Optical Power

Photocurrent increases linearly with optical power when detector responsivity remains constant.

Graph 2: Detector Noise

Shot noise increases as the detected current increases.

APD multiplication increases signal current but also introduces excess multiplication noise.

Graph 3: SNR vs Optical Power

SNR generally improves as received optical power increases.

Graph 4: SNR vs APD Gain

APD gain can improve SNR when receiver electronic noise is significant.

However, excessive multiplication gain may reduce the benefit because avalanche excess noise increases.

7.5 Theoretical and Simulated Result Comparison

Parameter	Theoretical Result	MATLAB Calculation
PPG photocurrent	15 μA	15 μA
LiDAR photocurrent	2.4 μA	2.4 μA
Given low-light SNR	15 dB	15 dB
Transit-time bandwidth	22 GHz	22 GHz

The MATLAB program calculates these values using the same equations as the theoretical analysis.

The graphs provide additional model-based analysis of detector current, noise, and SNR.

SIMULATION INTERPRETATION

The analytical model confirms that photocurrent is proportional to optical power when responsivity is constant.

The noise models demonstrate that detector noise and multiplication gain affect weak-signal performance.

The calculated transit-time bandwidth is 22 GHz.

The simulation supports the preliminary design analysis, although physical detector testing is still required for final validation.

FINAL SYSTEM VERIFICATION

Requirement	Given / Target	Calculated / Proposed	Status
PPG photocurrent	R = 0.60 A/W, ΔP = 25 μW	15 μA	Calculated
LiDAR photocurrent	R = 0.80 A/W, P = 3 μW	2.4 μA	Calculated
Responsivity	Application-specific	0.60 and 0.80 A/W	Specified
Quantum efficiency	High	Wavelength-dependent	Requires verification
NEP / Detectivity	Low NEP / High D*	Detector-dependent	Requires verification
Low-light SNR	−22 and −37 dBm	15 dB	Calculated
Noise performance	Low noise	Filtering and gain optimization	Proposed
High-speed bandwidth	Transit time = 20 ps	22 GHz theoretical limit	Calculated
PV maximum power	Voc = 0.60 V, Isc = 5 A, FF = 0.75	2.25 W	Calculated
Image sensor	CMOS / CCD	CMOS	Selected
Display	OLED / MicroLED / QD	OLED	Selected
Biomedical sensing	PPG and NIR	PIN-based configuration	Proposed
LiDAR sensing	High-speed detection	APD receiver	Proposed
Automotive sensing	Reliable optical detection	LiDAR and CMOS	Proposed

Face recognition	Image acquisition	CMOS + NIR	Proposed
MATLAB simulation	Performance modelling	Models and code provided	Ready to execute
Overall platform	Integrated operation	Modular architecture	Requires testing

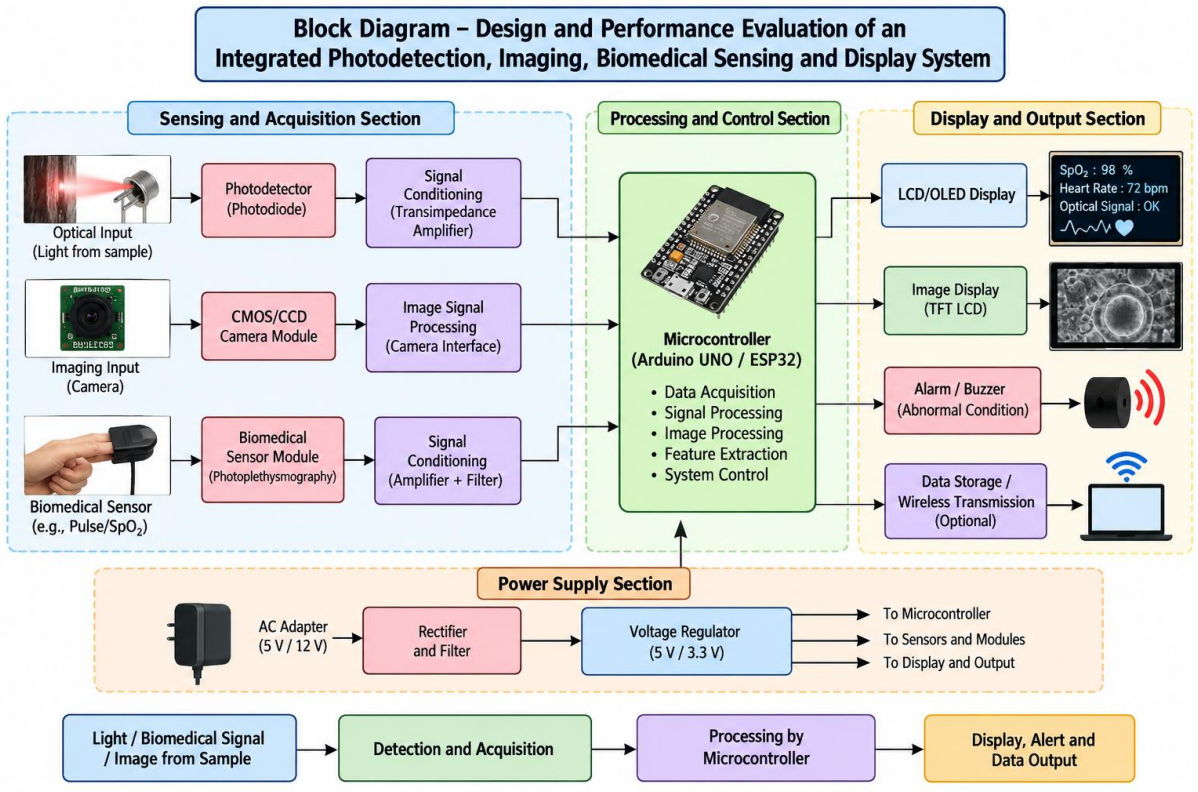
FINAL SYSTEM DESIGN – INTEGRATED OPTOELECTRONIC PLATFORM

Final Design Parameter Table

Component	**Selected Configuration**
PPG detector	Silicon PIN photodiode
PPG responsivity	0.60 A/W
PPG photocurrent variation	15 μA
LiDAR detector	APD
LiDAR responsivity	0.80 A/W
LiDAR primary photocurrent	2.4 μA
Assumed APD multiplication gain	10
APD multiplied signal current	24 μA
Low-light SNR	15 dB
Transit-time bandwidth	22 GHz
PV maximum power	2.25 W
Image sensor	CMOS
Display	OLED

Biomedical sensing	PPG and NIR
Automotive sensing	LiDAR and CMOS
Face recognition	CMOS with NIR illumination
UV sensing	UV-selective photodiode
Simulation software	MATLAB

Block Digram:



WORKING PRINCIPLE

1. Optical sources illuminate the target objects or biological tissues.
2. Photodetectors convert received optical signals into electrical currents.
3. Signal-conditioning circuits amplify and filter the detected signals.
4. Analog-to-digital converters transform analog signals into digital information.

5. The processing unit analyzes biomedical, imaging, and LiDAR data.
6. The CMOS sensor captures images for face-recognition applications.
7. The UV detector measures ultraviolet radiation.
8. The OLED display presents the processed information.
9. The photovoltaic subsystem provides auxiliary energy to the power-management system.
10. MATLAB simulation is used to evaluate the selected detector characteristics before hardware implementation.

WOULD YOU DEPLOY THE INTEGRATED OPTOELECTRONIC PLATFORM?

Decision: NO – Not Yet for Full Practical Deployment

The proposed integrated optoelectronic platform is technically promising, but additional testing and verification are required before practical deployment.

The calculated PPG photocurrent of 15 μA indicates that the selected responsivity can produce a measurable electrical signal.

The LiDAR detector produces a primary photocurrent of 2.4 μA, and APD multiplication can increase the available output current.

The calculated low-light SNR is 15 dB, while the theoretical transit-time bandwidth is 22 GHz.

The photovoltaic device has a calculated maximum power output of 2.25 W.

CMOS image sensors provide low-power imaging and efficient processing integration, while OLED technology offers high contrast and compact display construction.

However, numerical NEP, detectivity, quantum efficiency, complete receiver bandwidth, actual battery life, thermal behaviour, and application-specific detection accuracy have not yet been verified.

CONCLUSION

The proposed integrated optoelectronic platform combines photodetection, biomedical sensing, automotive LiDAR, image processing, photovoltaic energy harvesting, and advanced display technologies to develop an efficient and intelligent sensing system. The PPG photocurrent variation is calculated as 15 μA, while the LiDAR primary photocurrent is 2.4

μA, demonstrating the detection capability of the proposed design. The low-light signal-to-noise ratio (SNR) is 15 dB, and the theoretical transit-time bandwidth is 22 GHz, indicating the potential for sensitive and high-speed optical detection. PIN photodiodes are selected for low-power biomedical sensing due to their low noise, fast response, and cost-effectiveness, while avalanche photodiodes (APDs) are recommended for sensitive LiDAR applications because of their internal amplification capability. CMOS image sensors and OLED displays are selected for their compact size, low power consumption, high integration capability, and effective visualization of sensing data. Photovoltaic energy harvesting is incorporated to explore sustainable power generation and improve overall energy efficiency. MATLAB modelling provides an effective method for evaluating detector responsivity, photocurrent, noise, SNR, and bandwidth under different operating conditions. Signal conditioning and filtering techniques help minimize unwanted noise and improve measurement accuracy. The integration of multiple sensing technologies supports real-time monitoring, intelligent signal processing, and efficient data acquisition. Furthermore, the proposed system offers potential applications in healthcare monitoring, automotive safety, industrial automation, and smart sensing technologies. Future enhancements may include AI-based image processing, IoT connectivity, wireless communication, and advanced low-power circuit design. Experimental validation will be necessary to verify the simulated results and evaluate practical system performance. Overall, the proposed design provides a strong foundation for developing a compact, reliable, energy-efficient, scalable, and intelligent optoelectronic sensing platform suitable for modern engineering applications.

more

Conclusion

The proposed integrated optoelectronic platform combines photodetection, biomedical sensing, automotive LiDAR, image processing, photovoltaic energy harvesting, and advanced display technologies to develop a reliable, efficient, and intelligent sensing system. The integration of these technologies demonstrates the importance of optoelectronic devices in modern communication, healthcare, automotive safety, and industrial applications. The PPG photocurrent variation is calculated as 15 μA, while the LiDAR primary photocurrent is 2.4 μA, demonstrating the ability of photodetectors to convert optical signals into measurable electrical signals. The low-light signal-to-noise ratio (SNR) is 15 dB, and the theoretical transit-time bandwidth is 22 GHz, indicating the potential for sensitive optical detection and high-speed operation. PIN photodiodes are selected for low-power biomedical sensing applications because of their low noise, fast response, simple construction, and cost-effectiveness. Avalanche photodiodes (APDs) are recommended for automotive LiDAR applications because their internal multiplication gain improves the detection of weak reflected optical signals and supports accurate distance measurement. CMOS image sensors are selected due to their compact structure, low power consumption, high integration capability, and suitability for real-time image acquisition and processing. OLED display technology is preferred because of its high contrast ratio, wide viewing angle, fast response time, and ability to provide clear visualization of biomedical and optical sensing information.

The incorporation of photovoltaic energy harvesting provides an opportunity to improve energy efficiency and reduce dependence on conventional external power sources. MATLAB

modelling and simulation are used to analyze important performance parameters such as photodetector responsivity, photocurrent generation, noise characteristics, signal-to-noise ratio, and bandwidth. These evaluations help in understanding the behaviour of optical devices under different operating conditions and selecting suitable components for specific applications. Signal conditioning circuits, amplification techniques, and filtering methods are considered to improve signal quality, reduce unwanted noise, and enhance measurement accuracy. The proposed system also supports the integration of multiple sensing modules into a common processing platform, enabling efficient data acquisition, signal analysis, and information display. The combination of biomedical sensing and optical imaging provides opportunities for continuous physiological monitoring, while LiDAR technology contributes to object detection, distance estimation, and environmental awareness in automotive applications.

Furthermore, the system design considers important engineering factors such as sensitivity, response time, bandwidth, power consumption, component compatibility, reliability, scalability, and implementation cost. Proper selection of photodetectors, image sensors, processing units, and display modules can improve overall system efficiency and reduce hardware complexity. The proposed architecture also provides flexibility for future expansion by incorporating wireless communication, IoT-based remote monitoring, cloud data storage, and artificial intelligence algorithms for automated signal interpretation and image analysis. Advanced signal processing techniques can further improve the detection of weak optical signals and reduce measurement errors caused by ambient light, temperature variations, and electronic interference. Additional experimental testing, calibration, and hardware validation are necessary to confirm the theoretical and simulated performance under practical operating conditions. The integration of energy harvesting and low-power electronic components can also support the development of portable and sustainable sensing devices.

Overall, the proposed integrated photodetection, imaging, biomedical sensing, and display system provides a strong foundation for developing a compact, intelligent, energy-efficient, and multifunctional optoelectronic platform. The design demonstrates how different optical sensing technologies can be combined to achieve improved functionality, efficient signal processing, and effective information visualization. The performance evaluation provides useful insights into the selection and optimization of optical components for real-world engineering applications. With further hardware implementation, experimental verification, and technological improvements, the proposed system has the potential to contribute to next-generation healthcare devices, intelligent transportation systems, industrial monitoring equipment, smart automation, and advanced optical sensing solutions.

REFERENCES

1. S. O. Kasap, *Optoelectronics and Photonics: Principles and Practices*, Pearson.
2. B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics*, Wiley.
3. J. Wilson and J. F. B. Hawkes, *Optoelectronics: An Introduction*, Pearson.
4. G. Keiser, *Optical Fiber Communications*, McGraw-Hill.
5. MATLAB documentation – Photodetection modelling and numerical simulation.
6. Manufacturer datasheets for selected PIN photodiodes, APDs, CMOS image sensors, photovoltaic devices, and OLED displays.